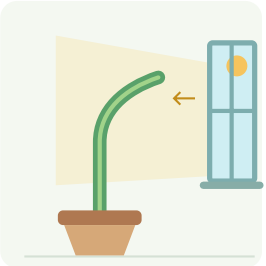
Reference answers

# Worksheet 3: Phototropic growth in plants

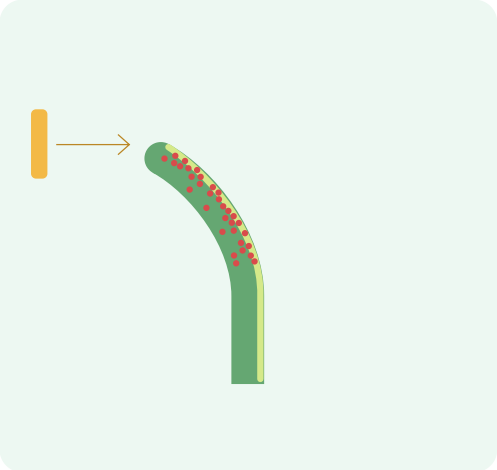
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## Topic notes

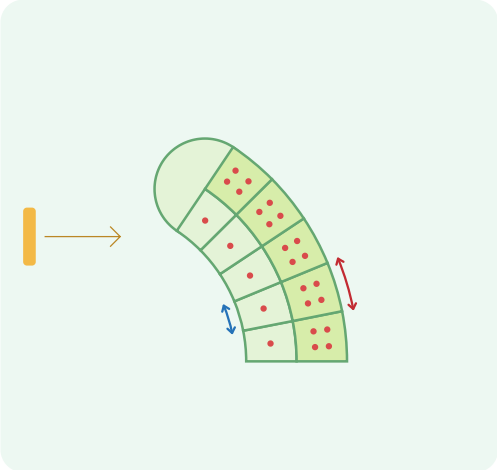


**Phototropism**（向光性） is growth in response to light direction; growth towards light is **positive phototropism**（向光性）. A coleoptile（胚芽鞘） protects young leaves.

The tip senses light. Its growth-promoting effect **passes down**. **Auxin**（生長素） is a **plant hormone**, not light or a nutrient.



Coleoptile（胚芽鞘）: more **auxin**（生長素） on the **shaded** right



Cell detail: more elongation（延長） on the right

Light on the left → **shaded** right side → more **auxin**（生長素） **passes down** → right-side cells **elongate**（延長） more → bending left.

Agar（瓊脂） supports a transferable growth-promoting effect, not its identity or distribution under light.

## Part 1 | Context and fair comparisons

Compare uncovered, tip-**shaded** and lower-**shaded** coleoptiles（胚芽鞘）. All tips are intact; other conditions stay the same.

Identify the variables: tick the appropriate factors in each row. You may choose more than one.

| Variable type | Factors to choose from |
|---|---|
| Independent variable (the factor changed on purpose) | □ Time exposed to light □ The coleoptile（胚芽鞘）'s bending response □ Light direction and intensity □ Coleoptile（胚芽鞘）type, starting height and growth stage ☑ **Light-blocking treatment** □ Temperature and water supply |
| Dependent variable (the result measured or observed) | □ Time exposed to light ☑ **The coleoptile（胚芽鞘）'s bending response** □ Light direction and intensity □ Coleoptile（胚芽鞘）type, starting height and growth stage □ Light-blocking treatment □ Temperature and water supply |
| Controlled variables (the factors kept the same) | ☑ **Time exposed to light** □ The coleoptile（胚芽鞘）'s bending response ☑ **Light direction and intensity** ☑ **Coleoptile（胚芽鞘） type, starting height and growth stage** □ Light-blocking treatment ☑ **Temperature and water supply** |

### What assumptions does this inquiry make?

An assumption is a condition we have not proved but take as true, to help make the experiment results valid.

Choose all suitable assumptions. You may choose more than one.

☑ **The coleoptiles（胚芽鞘） start in similar growth conditions.**

☑ **The light-blocking cap and sleeve do not limit elongation（延長） or bending.**

☑ **The treatments do not cause clear temperature differences between setups.**

□ If the plants are the same type, their starting size does not need to be controlled.

### Control group

To find out whether blocking light affects the coleoptile（胚芽鞘）'s growth direction, how would you design the control group?

(Hint: The control group provides a baseline for comparison. It helps you confirm that a change is caused by the independent variable rather than other external factors, so the control group does not receive the independent variable treatment.)

**Use intact, uncovered A as the baseline for B and C; keep all other conditions the same.**

## Part 2 | Results and analysis

### 2.1 Shading

Light is on the left. The sleeve covers below the exposed tip without restricting growth.

A · No light blocked B · Light blocked at the tip C · Light blocked below the tip



| Setup | Bending direction |
|---|---|
| A | （**Bends left** ／ Bends right ／ No clear bending） |
| B | （Bends left ／ Bends right ／ **No clear bending**） |
| C | （**Bends left** ／ Bends right ／ No clear bending） |

1. Compare A and B: after **blocking light at the tip**, what changes? （**No clear bending towards the light.** ／ It shows the same bending towards light.）

2. Compare A and C: when the lower part **does not** receive light directly, does the coleoptile（胚芽鞘） still bend? （**It can still bend towards light.** ／ No clear bending towards the light.）

3. Which part may **sense light from one side**? （**Tip** ／ Only the lower part）

### 2.2 Tip removal

A · No light blocked D · Tip removed



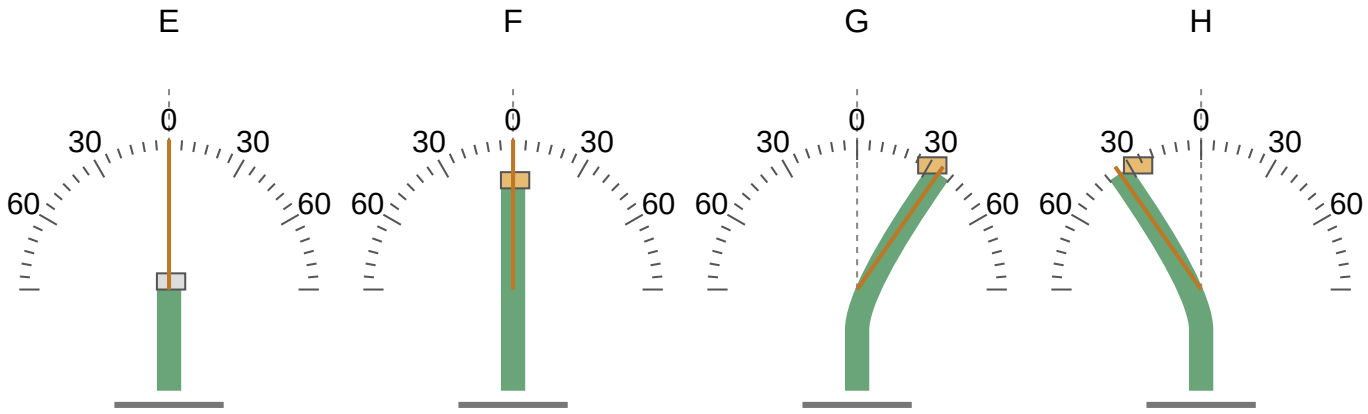
| Setup | Elongation（延長） |
|---|---|
| A | （**Yes** ／ No） |
| D | （Yes ／ **No**） |

1. What does comparing elongation（延長） in A and D support? （**The presence of the tip also affects the coleoptile（胚芽鞘）'s growth / elongation（延長）.** ／ The presence of the tip does not affect the coleoptile（胚芽鞘）'s growth / elongation（延長）.）

### 2.3 Agar（瓊脂）

Blank agar（瓊脂） has not touched a tip; treated agar（瓊脂） has. All cut-tip coleoptiles（胚芽鞘） stay in darkness; initial size, agar（瓊脂） size, time, temperature and water supply are the same.

E: blank agar（瓊脂）, centre; F: treated agar（瓊脂）, centre; G: treated agar（瓊脂）, left; H: treated agar（瓊脂）, right.




All start upright. Dashed lines mark vertical 0°; each small interval is 5°. E shows no clear elongation（延長）; F, G and H **elongate**（延長）. Angles and incubation are a teaching model.

| Setup | Angle (°) | Bending direction |
|---|---|---|
| E | 0° | （Bends left ／ Bends right ／ No clear bending） |
| F | 0° | （Bends left ／ Bends right ／ No clear bending） |
| G | 35° | （Bends left ／ Bends right ／ No clear bending） |
| H | 35° | （Bends left ／ Bends right ／ No clear bending） |

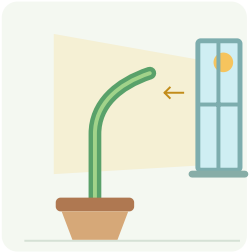
1. What does the comparison of E and F support? （The tip may produce a growth-promoting effect that can be transferred. ／ All agar（瓊脂） produces the same growth-promoting effect.）

2. What does the comparison of F, G and H support? （The position of treated agar（瓊脂） affects the direction of coleoptile（胚芽鞘） bending. ／ The position of treated agar（瓊脂） does not affect the direction of coleoptile（胚芽鞘） bending.）

3. When treated agar（瓊脂） is placed on the **left** side of a coleoptile（胚芽鞘） with its tip removed, compared with the right side, the left side receives （more ／ less） substance X, causing cells on the left to **elongate（延長）** （more ／ less） and the coleoptile（胚芽鞘） to bend to the （left ／ right） side.

## Part 3 | Integrated explanation

Use the experimental results and topic notes to explain why the coleoptile（胚芽鞘） in the diagram bends to the right.



Light is on the right and sensed at the tip. More auxin（生長素） is on the shaded left side and passes down. Left-side cells elongate（延長） more, causing bending towards the light on the right (positive phototropism（向光性）).